GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form A · 24 points · two class periods **ANSWER KEY**

Name
Date
Period

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors

10 points MA.7.GR.1.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 4 cm side of a figure is 10 cm in its scale copy. Find the scale factor. *1 pt*

k = 2.5

copy ÷ original

Scoring: 1 point.

2. A 15 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.4

copy ÷ original; the copy is smaller, so k is under 1 — 2.5, 4 and 2 are the proportion backwards

Scoring: 1 point.

3. A drawing has a 35° angle. A scale copy is made at k = 3. What is the matching angle in the copy? *1 pt*

- ○ A. 105°
- ● **B. 35°**
- ○ C. 11.67°
- ○ D. 38°

B

Lengths scale; angles do not.

Scoring: 1 point.

4. A 3 cm by 4 cm rectangle is scaled by k = 3.

a. Find the copy's perimeter. *1 pt*

42 cm

perimeter × k

b. Find the copy's area. *1 pt*

108 cm²

area × k² — 36, 40 and 30 multiplied the area by k once

Scoring: a 1 · b 1 = 2 points.

5. A figure has area 8 cm². Its scale copy has area 72 cm². Find k, and explain why k is not 9. *1 pt*

k = 3. The area ratio is k², because area is two lengths each multiplied by k; 9 is k², so k is its square root.

The point needs k AND the reason (two lengths, so the ratio is k²). 'k = 3' alone is half the item; 'because you square root it' with no reason why is the rule without the reason.

Scoring: 1 point.

6. A floor plan is drawn at ¼ in : 1 ft. A room is 2 in by 3.5 in on the plan.

a. Find the real length of the wall drawn 2 in long. *1 pt*

8 ft

the scale per 1 inch, then × — 1 in : 4 ft, 1 in : 2 ft, 1 in : 5 ft

b. Find the room's real area. *1 pt*

112 ft²

real dimensions first: 8 by 14, 6 by 10, 10 by 15 — the drawing's area × the rate once is the named error

Scoring: a 1 · b 1 = 2 points.

7. A plan is drawn at 1 in : 4 ft. A real wall is 18 ft long. How long is it on the plan? *1 pt*

4.5 in

real ÷ rate

Scoring: 1 point.

8. A plan is drawn at 1 in : 5 ft. A room is 2 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square metres. *1 pt*

About 9.30 m^2 (3.05 m by 3.05 m)

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. 10 ft by 10 ft; 3.05 × 3.05 = 9.3025. The same number by the other route: 100 ft^2 × 0.305^2 = 9.3025. 30.5 (100 × 0.305) used the ratio once on an area. A plan's area asked in metric appears on no review and in no question bank; its type was 5.10 board 8.

Scoring: 1 point.

Part 2. Converting Across Systems

8 points MA.7.AR.3.3

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimetres is 9 in? *1 pt*

22.86 cm

× the ratio whose 1 is inches

Scoring: 1 point.

10. 1 cm = 0.39 in. How many inches is 70 cm? *1 pt*

27.3 in (27.6 by ÷ 2.54)

× 0.39; the sheet's other ratio, ÷ 2.54, earns the point too when the work shows it. × 2.54 (177.8, 228.6, 279.4) is the inverted ratio.

Scoring: 1 point.

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimetres is 4 yd? *1 pt*

365.76 cm

a chain: 144 in, 180 in, 54 in, then × 2.54. The estimate (a yard is nearly a metre) catches 10.16, 12.7 and 3.81.

Scoring: 1 point.

12. 1 kg = 2.204 lb. How many pounds is 20 kg? *1 pt*

44.08 lb

× the ratio whose 1 is the unit you have; the estimate says more pounds than kilograms

Scoring: 1 point.

13. 1 gal = 3.785 L. How many litres is 5 gal? *1 pt*

18.925 L

more litres than gallons

Scoring: 1 point.

14. 1 yd = 3 ft. A rug is 4 yd^2 in area. How many square feet is that? *1 pt*

- ○ A. 12 ft^2
- ○ B. 9 ft^2
- ● **C. 36 ft^2**
- ○ D. 24 ft^2

C

Scoring: 1 point.

15. Rate on 5 Nov: 1 US dollar = 0.92 euros.

a. How many euros is $60? *1 pt*

55.20 euros

× the rate

b. How many US dollars is 23 euros? *1 pt*

$25

÷ the rate; 21.16, 72.90 and 87.12 multiplied both ways

Scoring: a 1 · b 1 = 2 points.

Part 3. Multi-Step Proportion Problems

6 points MA.7.AR.4.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car does 26 miles per gallon and its tank holds 15 gal.

a. How far does it go on a full tank? *1 pt*

390 mi

the rate × the gallons

b. It starts a 1,000-mi trip with a full tank. How many tanks of gas does the trip take? *1 pt*

3 tanks (1,000 ÷ 390 ≈ 2.56)

the part of a tank is still a tank; follow-through on part (a)

Scoring: a 1 · b 1 = 2 points.

17. A recipe for 4 servings uses 1½ cups of rice.

a. How much rice for 10 servings? *1 pt*

3¾ cups

k from the yields (2.5, 1.5, 3), then the ingredient

b. With 6 cups on hand, how many servings can be made? *1 pt*

16 servings

recipes the pantry makes (4, 4, 5) × the yield; 15, 24 and 25 scaled the yield by the cups

Scoring: a 1 · b 1 = 2 points.

18. Juice is mixed orange : mango : water = 3 : 2 : 4. A batch uses 9 cups of orange. How many cups is the whole batch? *1 pt*

27 cups

k from the part you know (3, 4, 4), then the whole (9, 8, 6 parts); 12, 16 and 8 found one other part and stopped

Scoring: 1 point.

19. 1 mi = 1.61 km. A 200-km drive. Car A averages 40 mi/h; car B averages 60 km/h. Which car gets there first, and by about how long? Explain. *1 pt*

Car A: 40 mi/h = 64.4 km/h > 60 km/h, so about 3.11 h against 3.33 h — about 13 minutes sooner.

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. The point needs the speeds in one unit, the choice, and the times (or the gap). 'Car B, 60 > 40' compared two units. A decision from two rates in different units appears on no review and in no question bank; its type was 5.10 board 9.

Scoring: 1 point.

Score Tracker

Benchmark	Questions	Possible	Earned
MA.7.GR.1.5	1, 2, 3, 4, 5, 6, 7, 8	10	
MA.7.AR.3.3	9, 10, 11, 12, 13, 14, 15	8	
MA.7.AR.4.5	16, 17, 18, 19	6	
Total		**24**	

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.